

Paper	ENCN342 – Fluid Mechanics and Hydraulics
Topic	Steady Open Channel Flow
Module	5 – Gradually Varied Flow

Summary

This module presents the concept of gradually varied flow. The analysis of free surface profiles along entire reaches is presented. This analysis is strongly dependent on the classification of channel slopes.

Prerequisites

Module 4

Objectives

At the conclusion of this module you should:

1. Be able to define gradually varied flow and be able to derive the differential equation that describes the variation of flow depth along a channel.
2. Understand and be able to explain the key principles upon which gradually varied flow analysis is based.
3. Be able to classify channel slopes.
4. Be able to deduce the standard free surface profiles for mild and steep sloped channels.
5. Be able to use the standard free surface profiles to analyse the free surface profile along a channel reach that may include a number of flow controls.
6. Understand how the discharge problem is handled.
7. Be able to implement either of the quantitative methods for computing free surface profiles.

References

Hunt 1995 pp 12.12-12.32

Key Points

1. Gradually varied flow

Gradually varied flow is the natural flow condition that occurs in open channels where the presence of friction, transitions, hydraulic structures and other flow controls interact to influence the shape of the free surface profile along a channel. Changes in flow depth are assumed to be “slow” along the channel.

2. Key principles

The analysis of gradually varied flows is based on the following key principles:

1. Friction must be considered for long reaches.
2. Uniform flow is the state a flow will tend towards if undisturbed.
3. Hydraulic structures and other flow controls draw the flow away from uniform conditions.
4. Flow is subcritical upstream of a flow control and supercritical downstream.
5. Resistance in non-uniform flows is modelled using uniform flow equations.

3. Governing equations

The equations governing gradually varied flow are

$$\frac{dE}{dx} = S_0 - S_f \quad (1)$$

where S_0 is the bed slope (positive when sloping downstream) and S_f is the total energy slope (again positive when sloping downstream). This can be written as

$$\frac{dy}{dx} = \frac{S_0 - S_f}{(1 - Fr^2)} \quad (2)$$

or, for a very wide rectangular channel, as

$$\frac{dy}{dx} = S_0 \frac{y^3 - y_0^3}{y^3 - y_c^3} \quad (3)$$

where y_0 is the uniform flow depth.

4. Classification of channel slopes

The slope of a channel is defined in the following way

$y_0 > y_c$	for mild slopes
$y_0 < y_c$	for steep slopes
$y_0 = y_c$	for critical slopes
$S_0 = 0$ and $y_0 \rightarrow \infty$	for horizontal slopes
$S_0 < 0$ and y_0 is undefined	for adverse slopes

For wide rectangular channels the ratio of the uniform and critical depths can be found from

$$\left(\frac{y_0}{y_c}\right)^3 = \frac{q^2}{C^2 S_0} \frac{g}{q^2} = \frac{g}{C^2 S_0} \quad (4)$$

5. Longitudinal surface profiles

Qualitative information and classification of longitudinal profiles can be obtained by analysing equations 1, 2, or 3. The standard surface profiles for mild, steep, critical, horizontal and adverse slopes are given in Figs 1, 2, 3, 4 and 5.

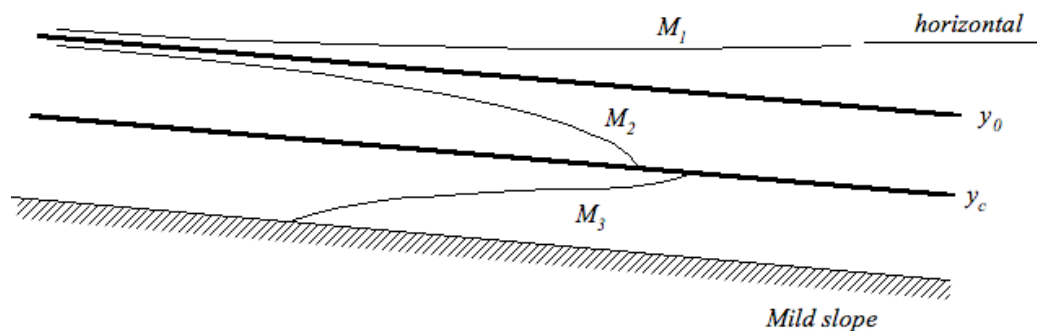
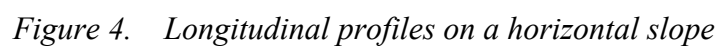
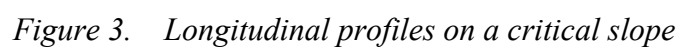
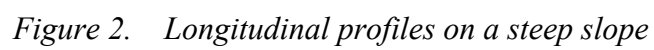


Figure 1. Longitudinal profiles on a mild slope



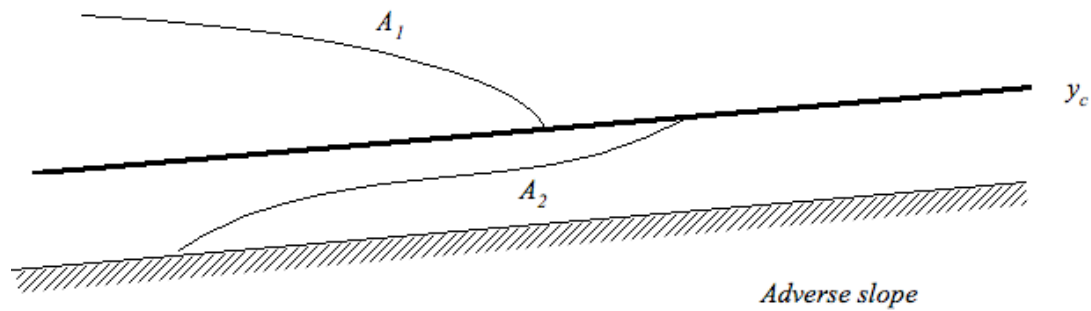


Figure 5. Longitudinal profiles on an adverse slope

6. Qualitative reach analysis

When analysing a reach that contains a number of flow controls and perhaps a number of different slopes and roughnesses a qualitative representation of the free surface profile can be obtained as follows:

1. Sketch the channel bed profile and add any hydraulic structures.
2. Calculate the uniform and critical flow depths and add these to the diagram.
3. Classify the slope along the channel.
4. Determine flow controls.
5. Work upstream and downstream of each flow control selecting the appropriate standard longitudinal profile.
6. Resolve conflicts between surface profiles, typically by inserting a hydraulic jump.

7. The discharge problem

The discharge problem arises when the discharge in a channel, typically originating in a lake or reservoir is not known. This is typically resolved by choosing a discharge, constructing the resultant free surface profile and checking whether the predicted specific energy at the upstream boundary matches that specified in the problem. If not the discharge is changed and the solution iterated until agreement is obtained. In the case of a mild slope downstream of the entrance to the channel the flow is uniform and the iterative process is rather trivial.

8. Quantitative profile calculation.

Quantitative predictions of free surface profiles flows the same method as above except in step 5 the governing ODE is integrated along the channel. If the equation is integrated for the depth the method is known as the **Standard Step Method**. If instead values for y are chosen and the equation is integrated for x then it is known as the **Direct Step Method**.

Worked Examples

Question 1

A very wide, rectangular cross-section channel is composed of 4 reaches the slopes of which, starting from the upstream end, are 0.001, 0.002, 0.005 and 0.002 respectively. The friction factor in each reach is 0.025, 0.030, 0.030 and 0.060. The channel emanates from a lake that has a water level of 6m above the channel bed at the exit point (you can take this as the specific energy at the upstream end of the channel). At the downstream end of the channel is a weir that controls the flow. A sluice gate, lowered into the flow, is located at the transition point between slopes 1 and 2. Qualitatively determine the free surface profile in the channel.

Solution

We will follow the method recommended in the notes to analyse this problem. However we note that the discharge is not known to begin with and we will need to compute that as part of the solution process.

Step 1. Fig 6 shows the channel bed profile (exaggerated of course) with the hydraulic structures and lake exit included.

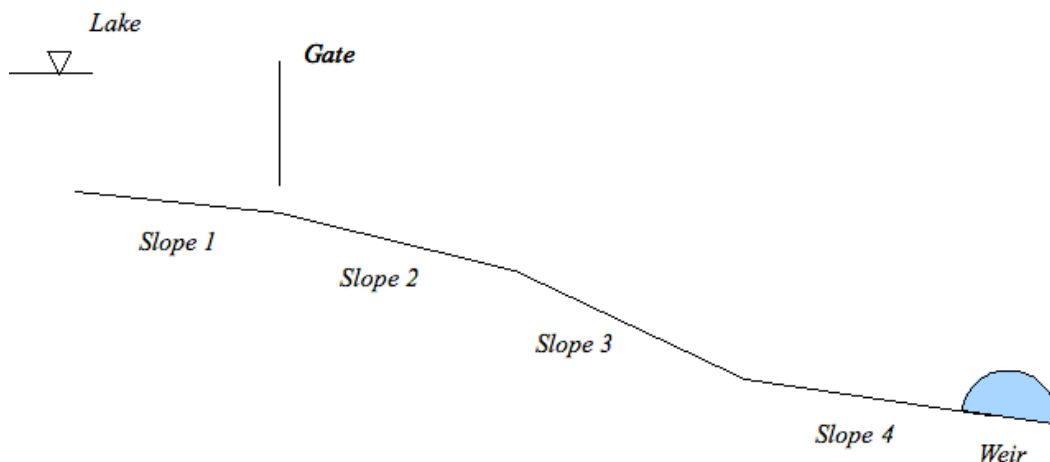


Figure 6. Schematic of the channel bed profile and hydraulic structures.

Step 2. The next step is to determine the critical and uniform depths. As the discharge is not known at this point we will use equation 4 to determine the ratio of the depths. This will determine whether the slopes are mild or steep. The analysis is given in table 1.

Table 1. The classification of the slopes using equation 4.

Slope	S_0	f	$C \text{ (m}^{1/2}\text{s}^{-1}\text{)}$	y_0/y_c	Class
1	0.001	0.025	56.0	1.46	Mild
2	0.002	0.030	51.1	1.23	Mild
3	0.005	0.030	51.1	0.91	Steep
4	0.002	0.060	36.2	1.55	Mild

The results in Table 1 indicate that all slopes are mild except slope 3.

At this point we can resolve the discharge problem by noting that the flow at the upstream end of the channel will be uniform and therefore we need to find a flow depth and flow velocity that both satisfies the appropriate resistance equation but also satisfies the specific energy condition at the entrance to the channel. We note that this analysis is only possible because uniform flow conditions must exist immediately downstream of the lake exit.

The two equations that must be satisfied are

$$q = C\sqrt{y^3 S_0} \quad (1)$$

and

$$E = 6 = y + \frac{q^2}{2gy^2} \quad (2)$$

where we have used the fact that the channel is very wide and rectangular so y can replace the hydraulic radius in the Chezy equation. These two equations can be easily solved for y by substituting equation 1 into equation 2. The result is

$$y = \frac{E}{1 + \frac{C^2 S_0}{2g}} = \frac{6}{1 + 0.16} = 5.17\text{m} \quad (3)$$

So finally the discharge is given by

$$q = C\sqrt{y^3 S_0} = 56\sqrt{5.17^3 \times 0.001} = 20.8\text{m}^2/\text{s} \quad (4)$$

With this information the uniform and critical flow depths on each slope can be computed. These are given in Table 2.

Table 2. The uniform and critical flow depths for the 4 slopes.

Slope	S_0	$C \text{ (m}^{1/2}\text{s}^{-1}\text{)}$	$y_0 \text{ (m)}$	$y_c \text{ (m)}$
1	0.001	56.0	5.17	3.53
2	0.002	51.1	4.36	3.53
3	0.005	51.1	3.21	3.53
4	0.002	36.2	5.49	3.53

Fig 7 includes these depths.

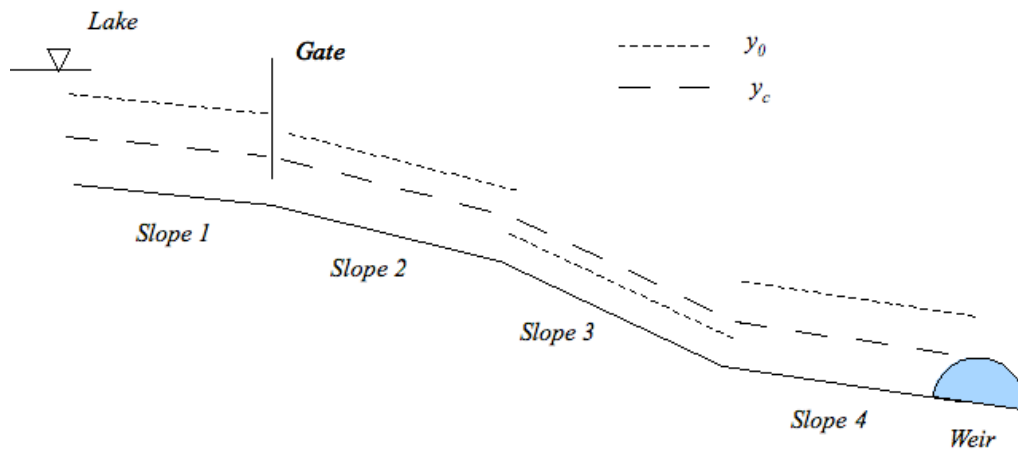


Figure 7. Schematic of the channel bed profile and hydraulic structures with the uniform and critical flow depths included.

Step 3. The slope classification has already been done in step 2.

Step 4. The points of control are guessed to be the following

- The sluice gate.
- The transition from slope 2 to slope 3 (a critical section).
- The downstream weir.

At the upstream boundary there is no critical section and the uniform flow on slope 1 could be considered as the flow control.

Step 5. Now we embark on the selection of free surface profiles upstream and downstream of each of these controls.

Sluice gate:

Upstream: The flow will be subcritical at the gate and tending to uniform flow far upstream towards the lake exit so this must be an M1 profile.

Downstream: The profile must be supercritical and thus must be M3.

Transition from slope 2 to slope 3:

Downstream: The flow will tend to uniform flow from critical flow so the profile is S2.

Upstream: The flow upstream will tend towards uniform flow and tend to critical at the transition so the profile is M2.

Downstream weir:

Downstream: The flow is supercritical but we are not interested in this portion of the flow.

Upstream: The flow will be subcritical and tending to uniform flow from critical at the weir so the profile is M2.

Step 6. Finally we must resolve any conflicts between these profiles. There are a number.

Downstream of the sluice gate the profile is M3 while upstream of the transition is M2. Assuming slope 2 is long enough the flow will return to uniform conditions downstream of the sluice gate and therefore a jump will occur which connects the M3 and M2 profiles.

Downstream of the transition from slopes 2 to 3 the flow is supercritical with an S2 profile. At the transition to slope 4 the flow will retain its supercritical nature and become an M3 curve on slope 4. As for the previous case this flow will ultimately jump to subcritical conditions provided the slope is long enough and the M3 profile will be connected to the M2 profile upstream of the weir with a hydraulic jump.

These profiles and jumps are illustrated in Fig 8.

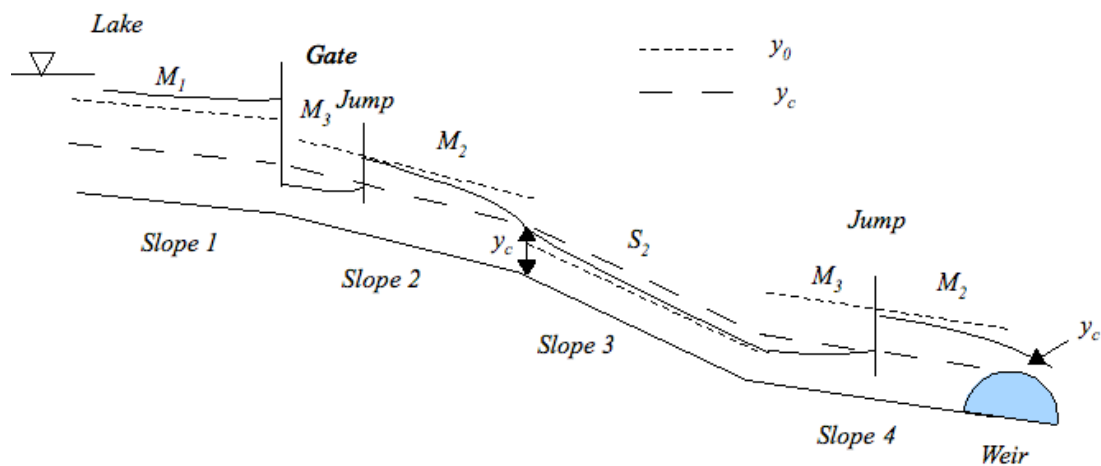


Figure 8. Schematic of the channel bed profile and hydraulic structures with the deduced free surface profiles included.

Question 2

Use the Standard Step Method to deduce the location of the hydraulic jump downstream of the sluice gate in question 1. You may assume that slope 2 is 2 km long and that the supercritical flow depth just downstream of the sluice gate is 1.2m. To find the jump location you will need to compute the conjugate depth for each of the flow depths in the M3 profile downstream of the gate.

Solution

This solution requires a numerical integration of the governing ODE as described in lectures. Matlab code to accomplish this is given below. It involves a script file that analyses the problem plus two functions used to integrate the ODE and find the conjugate depths using the iterative method.

```

%
% FreeSurfaceProfile
% =====
%
% This script implements a numerical integration of the governing equation
% for free surface profiles. For simplicity we analyse only problems for
% very wide channels with rectangular cross-section profiles.
%
% Inputs are:
%   C - Chezy C
%   q - 2D discharge
%   S0 - bed slope
%   deltaX - distance between nodes along the channel.
%   N - number of spatial node.
%
% The problem considered is the flow on slope 2 in question 2 of the
% worked solutions in Open Channel Flow Module 5.
% 5 Summary
%
% All units are SI.
%
deltaX = 20;
N = 101;
C = 51.1;
q = 20.8;
S0 = 0.002;
g = 9.81;
%
% Initialise the working variables variables.
%
yc = (q^2/g)^(1.0/3.0);
y0 = yc*(g/(C^2*S0))^(1.0/3.0);

%
% We solve this problem by integrating the governing ODE
% upstream of the critical section and downstream of the sluice gate.
%
% x1 and x2 are vectors that contain the x coordinates along the channel.
% x1 starts at the sluice gate and x2 starts at the critical section. The
% depths used to start the two integrations are 1.2m at the upstream end
% and a depth that is 1% greater than critical at the downstream end.
%
% The upstream integration is straightforward as the flow changes from
% near critical to uniform. However the downstream integration will
% have problems as the flow tends to critical flow and will produce
% spurious results further downstream.
%
% Once the downstream integration has been completed the conjugate depth
% for each depth is calculated the point at which the conjugate depth
% matches the upstream integration depth is used to identify the location
% of the jump. At this point the data for the upstream integration is
% truncated so that it stops at the jump.
%
% First we set up our vectors and initialise them.

x1 = zeros(N,1);
x2 = zeros(N,1);
yInitial = zeros(1,1);

for i = 1:N
    x1(i,1) = (i-1)*deltaX;
    x2(i,1) = 2000 - (i-1)*deltaX;
end
%
% We now integrate downstream from the gate
%
yInitial(1,1) = 1.2;
[X1,Y1] = IntegrateODE(S0,y0,yc,x1,yInitial);
%

```

```

% We now integrate upstream from the critical section
%
yInitial(1,1) = 1.01*yc;
[X2,Y2] = IntegrateODE(S0,y0,yc,x2,yInitial);
%
% We now compute the conjugate depths and store them in Y3
%
Y3 = Y1;
for i = 1:N
    Y3(i) = FindConjugateDepth(Y1(i),yc);
end
%
% Now we look for the point where the difference between Y3 and Y2
% changes sign - we will never be lucky enough to exactly find the point
% where they are the same.
%
diff = Y3(1) - Y2(N);
i = 1;
while diff > 0 && i <= N
    i = i+1;
    diff = Y3(i) - Y2(N-i);
end
X4 = zeros(i,1);
Y4 = zeros(i,1);
Y5 = zeros(i,1);
for j = 1: i
    X4(j) = X1(j);
    Y4(j) = Y1(j);
    Y5(j) = Y3(j);
end
%
% Finally we plot the three profiles:
%
% Y2 is the upstream profile
% Y4 is the downstream profile up to the jump
% Y5 is the conjugate depth profile up to the jump
%
plot(X4,Y4,X2,Y2,X4,Y5);

xlabel('x (m)');
ylabel('y (m)');
legend('Curve from sluice gate', 'Curve from critical section', 'Conjugate
depths', 'Location', 'SouthEast');

function [X,Y] = IntegrateODE(S0,y0,yc,x,yInitial)
%
% This function integrates the free surface profile.
%
[X,Y] = ode45(@dydx,x,yInitial);

function rhs = dydx(x,y)
rhs = S0*(y(1)^3 - y0^3)/(y(1)^3 - yc^3);
end
end

function y2 = FindConjugateDepth( y1,yc )
%
% This function finds the conjugate depth to y1, given the critical depth
% yc.
%
yp = y1/yc;
Mp = 1/yp+1/2.0*yp^2;
if yp < 1
    yp2 = 1.2;
    error = 1;
    while error > 0.001

```

```

        yp2Old = yp2;
        yp2 = sqrt(2*(Mp-1/yp2Old));
        error = abs(yp2 - yp2Old);
    end
end

if yp > 1
    yp2 = 0.8;
    error = 1;
    while error > 0.001
        yp2Old = yp2;
        yp2 = 1.0/(Mp - 1.0/2.0*yp2Old^2);
        error = abs(yp2 - yp2Old);
    end
end

if yp == 1
    yp2 = 1;
end

y2 = yc*yp2;

end

```

Fig 9 shows the resulting free surface profile with the location of the jump identified at approximately 400m downstream of the sluice gate.

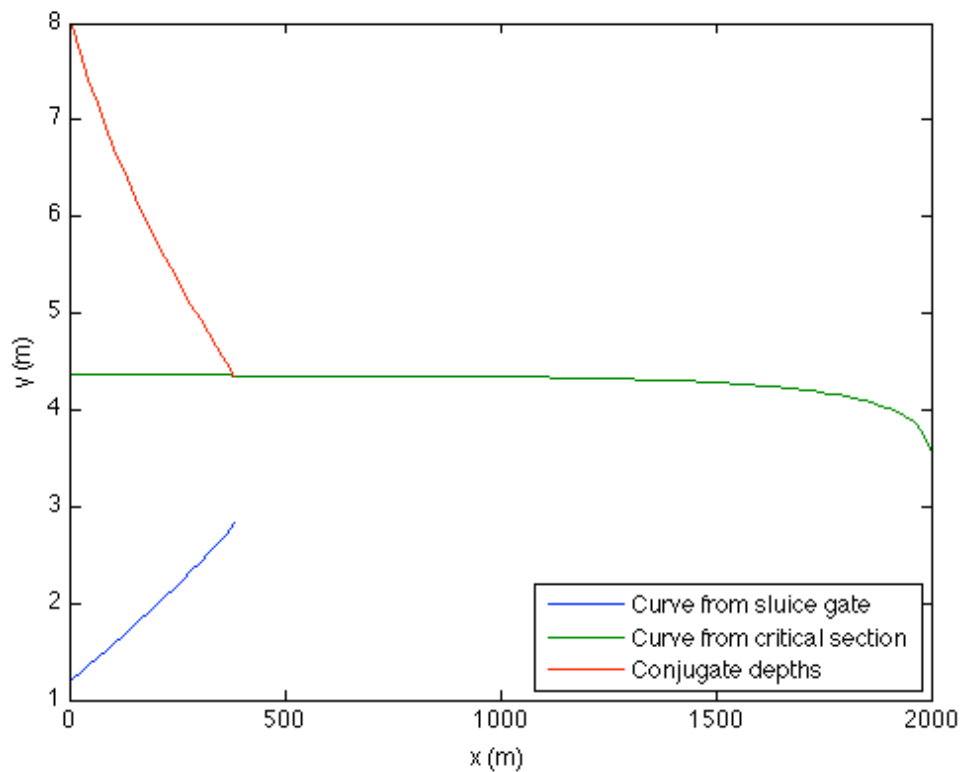


Figure 9. The free surface profile between the sluice gate and critical section. The point of discontinuity at 400m corresponds to the location of the hydraulic jump.